

Income tax — input to the renewable-energy investment decision

Additional power purchase agreements versus direct investment in generation capacity. Prepared to slot into the "Incremental cash flows + NPV" section, in place of the "Tax calculation" placeholder. Rates and sections are drawn from the income tax research on Document 7, pages 18 to 20, and from SARS Guide IT39 dated 23 November 2024.

A. How tax enters the model

- Match the tax basis to the discount rate. If the cash flows are pre-tax, discount at a pre-tax rate and leave tax out entirely. If they are post-tax, discount at a post-tax WACC and apply everything below. Mixing the two is the most common error in this calculation.
- Corporate rate 27%; capital gains inclusion rate for a company 80%, giving an effective 21.6%.
- Capital expenditure is a cash flow but not a deduction. Relief comes only through the capital allowance sections, so the allowance schedule is a separate line in the model.
- Capital allowances are non-cash. Bring them in as a tax saving (allowance x 27%), not as a cost.
- Only the direct investment generates capital allowances. That is the whole of the tax difference between the two alternatives.

B. The tax position of each option

Tax dimension	Additional PPAs	Direct investment	Why it matters to the decision
Capital allowance	Nil. s12B, s12BA and s11(e) each require the asset to be owned by the taxpayer and brought into use in that taxpayer's own trade. The producer owns the plant.	s12B(1)(h): 50 / 30 / 20 over three years. s12B(2)(b): 100% in year one for photovoltaic solar not exceeding 1MW. s12B(5) caps total deductions at cost.	The entire tax differential between the two alternatives sits in this row.
Allowance without ownership	Available only if the arrangement is in substance an instalment credit agreement, in which case s12B is available to Axion despite the producer holding title.	s12N deems a holder of a right of use to be the owner of improvements, but only for land or buildings leased from government or held under a public private partnership.	A PPA structured as an instalment sale changes the answer entirely. Test the legal form of the contract before concluding that no allowance is available.
Operating deduction	s11(a) read with s23(g): the full cost of electricity, every year, no rate and no write-off period. Not of a capital nature because it works the income-earning operations rather than adding to the structure.	s11(a) on operating, maintenance and insurance costs, and on wheeling and network charges.	Both options give a full deduction for operating cost. Neither is advantaged here.
Timing	The year the unconditional liability arises. Matches the cash cost, so there is no timing advantage or disadvantage.	Year of first use, with no apportionment for s12B even if first use falls on the last day of the year. s11(e) is apportioned. s12U(3) and s11A pre-trade costs fall in the year trade commences and cannot be postponed.	Front-loading is the source of the NPV benefit of owning. See section D.
Absorption of the deduction	Always absorbed, because it is a cost against operating profit.	s20(1)(a): a balance of assessed loss brought forward is set off only against the higher of R1 million or 80% of taxable income. s24H(3) limits a partner's share to the at-risk amount, with the excess carried forward under s24H(4).	A deduction that cannot be used in the year it arises is worth materially less. This is the largest single variable in the tax comparison. See section E.
Funding cost	Not applicable. Document 7 records that a PPA may require limited upfront capital.	s24J: interest is deducted as incurred on a yield-to-maturity basis, even where capitalised into the cost of the asset for	Bank debt is fully deductible. Shareholder, technical partner or offshore funding defers the deduction and

		accounting. s23M caps the deduction where the lender is in a controlling relationship with the borrower and is not subject to South African tax on the interest; the excess is carried forward, not forfeited.	changes the post-tax cost of debt in the WACC.
Disposal or exit	Nothing to dispose of.	s8(4)(a): recoupment equals proceeds limited to cost, less tax value, capped at allowances previously granted. Tax value is nil once the s12B allowances are exhausted. s8(4)(k) substitutes market value on a disposal to a connected person. s11(o) is available only where the tax life does not exceed 10 years and the buyer is not a connected person; it does not apply to s12U assets. Proceeds above cost give a capital gain at 80%.	Owning creates a tax liability on exit. It belongs in the final year of a finite-life model, not in the terminal value.
VAT	Input tax on the producer's invoice is deductible in full, because Axion makes taxable supplies. The producer must be a registered vendor.	Input tax of 15% on plant and construction cost is recoverable in the tax period of acquisition on a valid tax invoice. Import VAT is recoverable; customs duty is not, and forms part of the cost of the asset for s12B.	VAT is cash flow timing only in both cases, except for the customs duty, which increases the s12B base.
Carbon tax	Not imposed on purchased electricity.	Not imposed on generated renewable electricity.	Axion's carbon tax liability arises from the diesel backup generators. Document 7 page 20 records that renewable generation does not replace resilience infrastructure, so this cost is unchanged by either alternative and is not incremental.

C. Building tax into the incremental cash flows

This expands the taxable income build-up already in the model. The final column states where each line belongs, so the tax computation and the cash flow do not double count.

Cash flow line	Additional PPAs	Direct investment	Treatment in the model
Energy benefit — grid electricity avoided	Taxable	Taxable	Include at x (1 - 27%)
PPA payments for electricity	s11(a), deductible in full	Not applicable	Include at x (1 - 27%)
Wheeling, network and administration charges	s11(a)	s11(a), where generation is off site	Include at x (1 - 27%)
Operating, maintenance and insurance	Not applicable	s11(a)	Include at x (1 - 27%)
Surplus electricity sold, wheeled or credited	Gross income when it accrues	Gross income when it accrues	Include at x (1 - 27%), and only where the contract permits the sale
Amounts paid in advance under the contract	s23H apportions the deduction over the period to which the expenditure relates	s23H, if any amount is prepaid	Cash out in full in the year paid; deduction spread. A timing difference only
Initial capital expenditure	Not applicable	Not deductible	Cash out in full at CF0. No tax effect on this line
Capital allowances	Nil	s12B, s12U, s13quin and s11(e) per section D	Non-cash. Include as a tax saving of

			allowance x 27%
Land	Not applicable	No allowance under any section	Cash out in full. Relief only as base cost on a later disposal
Replacement capital expenditure	Not applicable	A qualifying replacement asset starts a fresh s12B 50 / 30 / 20. The asset replaced triggers s8(4)(a) or s11(o)	Cash out in full, with its own allowance schedule from the year of first use
Interest and funding cash flows	Not applicable	s24J, subject to s23M	Exclude from project NPV where the discount rate already reflects financing
Working capital movement and recovery	No tax effect	No tax effect	Cash flow only
Disposal at the end of a finite life	Not applicable	Proceeds are not themselves taxable. The recoupment under s8(4)(a) and any capital gain are	Final year cash flow. Exclude from terminal value, consistent with the treatment of once-off items

D. Capital allowance schedule

Illustrative, using the costs in the income tax research. Replace with the figures given on the day.

Cost element	Cost R'm	Section	Basis
Generation plant, 30MW photovoltaic	450.0	s12B(1)(h)	50 / 30 / 20, because capacity exceeds 1MW
New road and perimeter fencing	25.0	s12U(1)(a)	100% in year one, if the trade of generating electricity exceeding 5MW is carried on
Grid connection and transmission line	60.0	s11(e)	Written off over the IN 47 period, apportioned for part-periods of use. 20 years assumed; confirm on the day
Land	15.0	None	No allowance. Base cost only
Total	550.0		

Resulting deductions and tax savings

Year	s12B plant	s12U road and fence	s11(e) grid	Total deduction	Tax saved at 27%
1	225.0	25.0	0.5	250.5	67.6
2	135.0	—	3.0	138.0	37.3
3	90.0	—	3.0	93.0	25.1
4 to 20	—	—	3.0 a year	3.0 a year	0.8 a year
21	—	—	2.5	2.5	0.7
Total	450.0	25.0	60.0	535.0	144.4

All amounts R'million. 90% of the total deduction falls in years 1 to 3, which is where the value of owning the plant comes from. Discounted, the R144.4 million of tax saved is worth roughly R116 million at 10%, R110 million at 13% and R105 million at 16%. Use the project WACC, not these rates.

E. Who absorbs the deduction — the structuring choice

The allowance rate is identical in all three columns. What differs is whose taxable income absorbs it. Document 7 page 20 records that the investment may be made with an experienced technical partner but does not state the structure, so this must be established on the day.

	Axion owns the plant	Joint operation, 50% share	Separate project company
Who claims	Axion	Axion 50%, partner 50%	The company
Year one allowance to Axion	R250.5m	R125.25m, subject to the s24H(3) at-risk limit	Nil. Axion deducts only s11(a) on electricity bought
Tax saved in year one	R67.6m	R33.8m, if the at-risk amount is sufficient	Nil to Axion
Absorbed against	Taxable income of about R350m, so fully used	Partly used; the excess is carried forward under s24H(4)	The company's own income only

Effect of s20(1)(a)	None	Possible, on any resulting assessed loss	Severe. The allowances create an assessed loss in a company with little other income, and the 80% limitation means tax is still paid on 20% of taxable income above R1m while a large loss is carried forward
If the plant accedes to land Axion does not own	Nil to Axion, and nil to the landowner, who incurred no cost	Nil to Axion	Unaffected, provided the company owns the plant

The conclusion this drives: holding the plant where there is existing taxable income to shelter converts a deferred deduction into an immediate one. On these figures that is worth the difference between R67.6 million of tax saved in year one and nil. Structure is worth more than the rate.

F. Tax assumptions and the risk if they are wrong

Assumption	Risk if the assumption is incorrect	Impact
Axion owns the generation plant and brings it into use for the first time in its own trade.	The plant is owned by the technical partner or a project company, or it accedes to land Axion does not own under superficies solo cedit.	No s12B allowance to Axion at all. Where the plant accedes to land owned by another party, there is also no deduction to the landowner, who incurred no cost. The entire R144.4 million tax benefit falls away.
Each generating asset draws only on a specified renewable source.	An individual asset draws on both renewable and non-renewable sources, for example a hybrid inverter, a shared direct current bus, or a battery charged from both the plant and the diesel generators.	That asset qualifies for no deduction at all, with no apportionment. The risk attaches to the individual asset, not to the site, so separate diesel generators elsewhere on site do not taint a standalone renewable plant.
The generation trade test in s12U is met.	Generation is solely for Axion's own consumption and no trade of generating electricity is carried on, or capacity does not exceed 5MW.	The 100% deduction on roads and fencing is lost and that expenditure falls outside every allowance section. On the illustrative figures, R25 million of year-one deduction and R6.8 million of tax.
Axion has sufficient taxable income in years one to three to absorb the allowances.	The plant is held in a structure with little other income, or Axion's own taxable income falls.	An assessed loss arises and s20(1)(a) restricts its use to the higher of R1 million or 80% of later taxable income. The tax benefit is deferred over several years and its present value falls sharply.
The plant is funded by a lender that is subject to South African tax on the interest.	Funding is provided by a shareholder, the technical partner or an offshore group company in a controlling relationship that is not subject to South African tax on the interest.	s23M caps the interest deduction by reference to adjusted taxable income. The excess is carried forward rather than forfeited, so the effect is deferral, but the post-tax cost of debt in the WACC rises.
Assets outside the generation system are correctly classified.	Transmission lines, transformers and grid-charged batteries are treated as qualifying for s12B.	These assets do not generate electricity and generally fall to s11(e) over a much longer write-off period. Classifying them as s12B overstates early deductions and the NPV.
The write-off period used for s11(e) assets is correct.	The IN 47 period for the transmission line and related assets differs from the 20 years assumed.	A modelling error only, but it shifts the timing of a R60 million deduction.

G. Read-across to the decision

	Additional PPAs	Direct investment
Value	Tax scales the cost and nothing more. A tariff of R1.10 per kWh is R0.803 per kWh after tax. No present value gain, no present value cost.	Front-loaded allowances create a genuine present value gain, because 90% of the deduction falls in years one to three. Against this, the recoupment on exit is a real cash cost in the final year.
Risk	Low tax risk. The only real exposure is that the producer must be a registered vendor for the input tax to be deductible.	Four conditions each carry the whole allowance: ownership, asset-level source purity, the s12U trade test, and sufficient taxable income to absorb the deduction. Each

		is binary.
Strategic	Predictable and stable tax position over a contract of three to five years. No reliance on an incentive that may change.	The s12BA window, which gave 125% in the year of first use, closed on 28 February 2025. Plant brought into use now receives 50 / 30 / 20. The tax case for owning is weaker than it was two years ago and must not be built on s12BA. Separately, confirm whether the existing on-site solar plant was first brought into use between 1 March 2023 and 28 February 2025, which would have entitled Axion to 125% of its cost in that year.
Capital allocation	No capital required, so no capital is diverted from the remaining 5MW of saleable IT capacity or the undeveloped land.	Of the R550 million illustrated, R15 million of land attracts no allowance under any section. The tax benefit of the remaining R535 million is realised early only if the plant sits where there is taxable income to absorb it.